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CAR DAMAGE A.I

Intro Artificial Intelligence

CAP 4630

An automated classification & Severity Scoring of Vehicle damage

OVERVIEW

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Results & Demo

Conclusion

Objectives

PROBLEM

Manually assessing car damage is often slow, inconsistent, and very costly. And for some there may be visible damages that are not critical to safety but drivers dont hesitate to fix.

GOAL

Create an AI system that classifies vehicle damage types and assigns an actionable severity score (1-5) from a provided image.

IMPACT

Allows for faster claims processing, objective damage evaluation, and scalable deployment in insurance, rental fleets, and repair shops.

AI models & Approach

Classical ML(primary)

Soft-voting Ensemble:

- Random forest
- Support Vector Machine
- HistGradientBoosting

Why Ensemble? It combines complementary strengths - RF handles noisy features, SVM finds decision boundaries, GBM correct residual Errors.

DATASET

Source: Kaggle Vehicle Damage Dataset

Images : 224 x 224 px, RGB

Split: 80% Train | 20% Test | 15% Val

Format: CSV with image filenames &class

DAMAGE CLASSES:

- Bumper-Scratch
- Door- scratch
- Bumper-Dent
- Door-Dent
- Head-Lamp
- Tail-Lamp
- Glass Shatter
- Unknown

METHODOLOGY & PIPELINE

1 Data Loading → Read CSV metadata, locate images, skip the unknowns if configured

2 Preprocessing → Resize images to 224x224, normalize pixels to [0,1] float32

3 Feature Extraction → HOG + Color Histogram

4 Train/Test Split → 80/20 split , separate 15% validation set for DL

5 Model Training → Soft-voting ensemble fit on combined HOG + Histogram Features

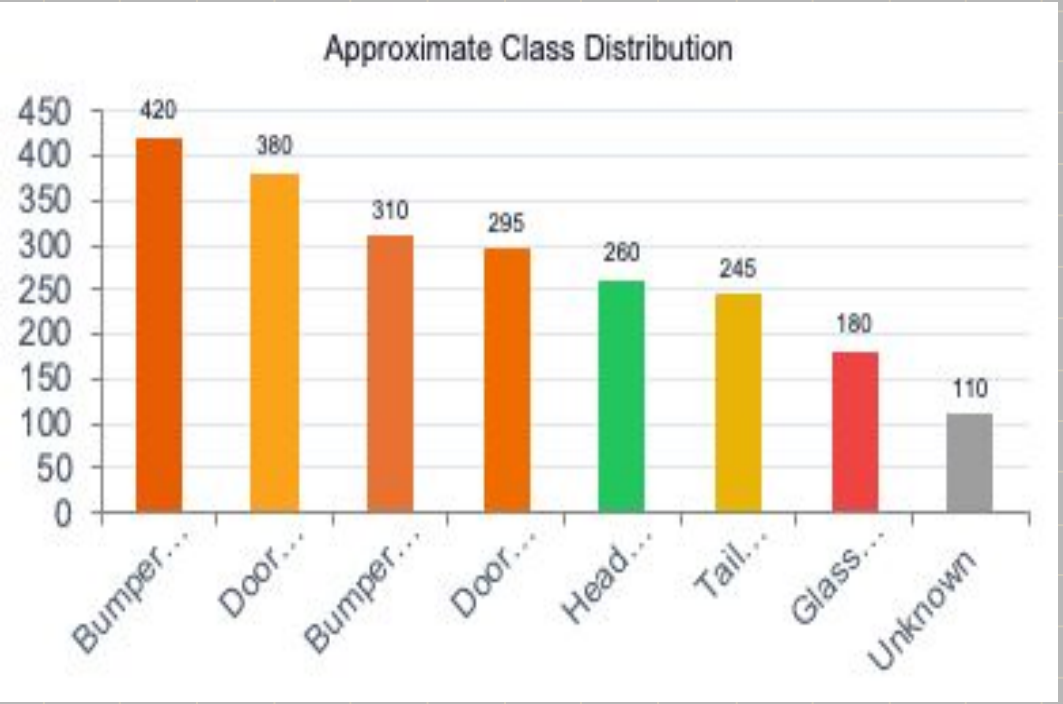
6 Severity Scoring → Predicted class -> BASE_SEVERITY map (1-5) adjusted by confidence

7 Evaluation → Accuracy, Precision, Recall, F1 Score, Confusion Matrix

Pred: bumper_dent
Score: \mathbf{3}/5

Severity Scoring system

1. UNKNOWN - no significant damaged detected.
2. LOW - Cosmetic damage only(scratch/scuff)
3. MEDIUM-Structural damage(repairs advised)
4. HIGH - Safety Concerns(urgent repairs needed)
5. CRITICAL - Vehicle unsafe to drive(immediate action needed)



Training Accuracy ~ 82 %

Validation Accuracy ~ 74 %

Test Accuracy ~ 71 %

Weighted F1 Score ~ 0.70 %

Live Demo - run_enhanced_demo()

Each predicted image received a colored border based on severity:

GREEN= Score 1-2 (minor) **YELLOW** = Score 3(Moderate) **RED** = Score 4 - 5(critical)

Challenges and lessons learned

Challenge

Dataset Contained 'unknonw' class and inconsistent label casing('head_lamp') Solution: lowercase normalization + skip_unknown flag

File path & Module Structure

I used colab notebook creating most of this project and it required careful sys.path management and importlib.reload() calls to make sure updated modules were used during iterative development

Feature Vector Design

Early models using raw pixel values performed poorly. HOG + Color histogram combination helped improved class seiparability across severity types.

Ensemble > Single Model

Soft voting combined random forest's robustness, SVM's boundary precision, and GBM's error correction outperforming any single classifier alone.

Lesson: Severity as business value

A classification label alone is insufficient for real world usage. Mapping classes to a 1-5 severity scale makes the output directly actionable for repair triage

Conclusion....

Achievements

- Multi-class classifier with HOG + Color Histogram features
- Soft-voting Ensemble for robust predictions
- Custom severity scoring linked to damaged classes

Limitations

- Accuracy drops on ambiguous or occluded damage regions
- 'Unknown' class is difficult to distinguish from minor damage
- Model generalizes to dataset domain, real world variation is broader

Future Work

- Fine-tune for higher accuracy
- Collect more balanced labeld data across a wider range of data types
- Deploy as web/mobile API for real time usage

THANK YOU!

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